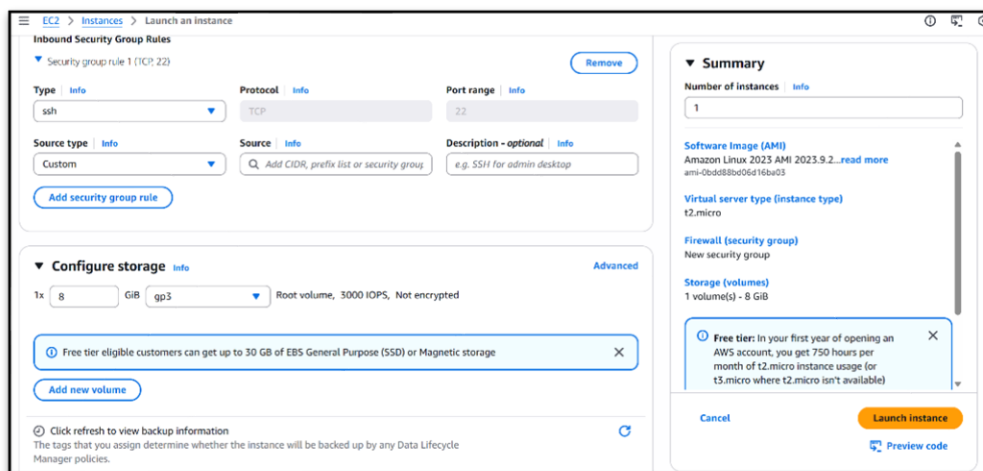


ASSIGNMENT 1

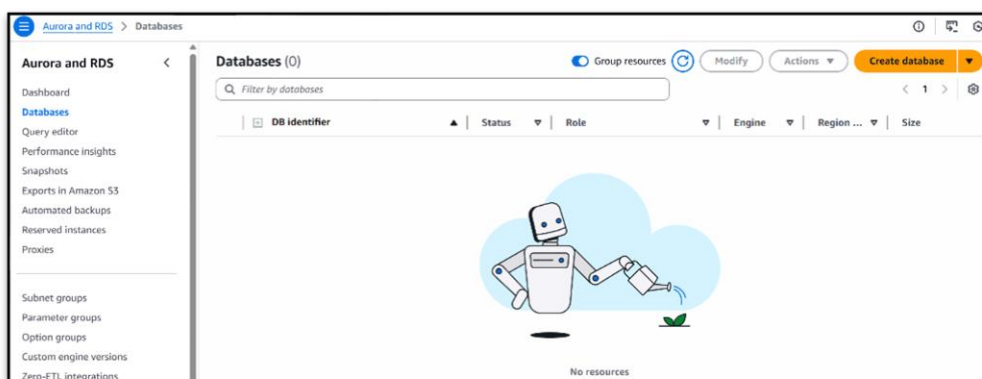
Task 1:

To create a MySQL DB instance and to connect it with the EC2 instance to create and read the queries executed using the MySQL commands.

- Open the Amazon EC2 → Click on launch instance → Name → Select the key pair → Allow ssh port 22 on the security inbound rules to securely connect with the remote device → Launch instance.



- The instance has been successfully launched.
- Go to the Aurora and RDS services by navigating through the search bar → Click on databases → Create database → Standard create → Choose MySQL engine → Free tier → Type a master username or leave it as default(admin) → Type a master password (remember it for the connectivity in the linux server) → Choose a VPC or create one and remember the security group name.



Create database Info

Choose a database creation method

☒ **Standard create**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type Info

☐ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☒ **MySQL**

☐ PostgreSQL

☐ MariaDB

☐ Oracle

Additional configuration
Database options, encryption turned on, backup turned off, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Estimated monthly costs
The Amazon RDS Free Tier is available to you for 12 months. Each calendar month, the free tier will allow you to use the Amazon RDS resources listed below for free:

- 750 hrs of Amazon RDS in a Single-AZ db.t2.micro, db.t3.micro or db.t4g.micro Instance.
- 20 GB of General Purpose Storage (SSD).
- 20 GB for automated backup storage and any user-initiated DB Snapshots.

[Learn more about AWS Free Tier.](#)

When your free usage expires or if your application use exceeds the free usage tiers, you simply pay standard, pay-as-you-go service rates as described in the [Amazon RDS Pricing page](#).

You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

[Cancel](#) [Create database](#)

- The database has been successfully created.
- Click on the database created → VPC security group → Select the security group you have given when connecting with database → Edit inbound rules → Type: MySQL/Aurora → In the source select your EC2 Security group or enter its private IP range.

Edit inbound rules Info

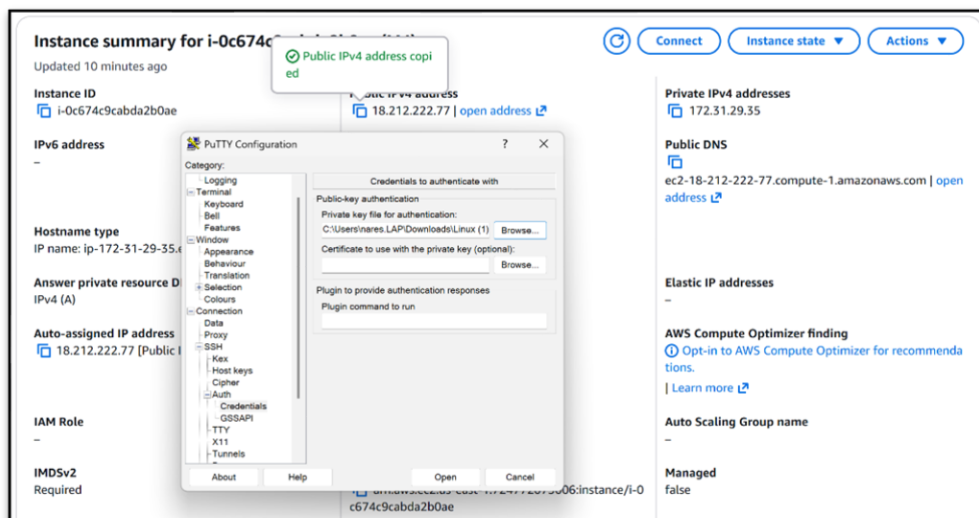
Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sg-0d093744b9a4b230f	MySQL/Aurora	TCP	3306	Custom 172.31.29.35/32	

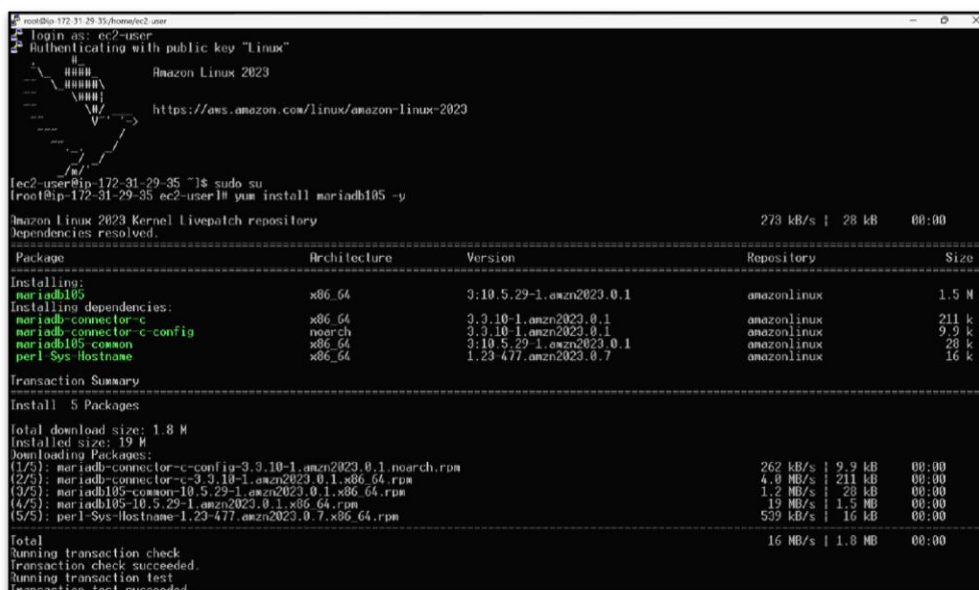
[Add rule](#) [Delete](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

- Open Putty or any other such as MobaXterm → In the host name enter the IP of the EC2 instance → SSH → Auth → Credentials → Browse and provide the private key.



- The Connection has been established.
- Login as ec2-user → sudo su (super user) → yum install mariadb105 -y to install MySQL and yum update to update MySQL → mysql -h endpoint of database -u master username -p → Enter the master password (To login into MySQL).



- Create database → Use database → Create table → Insert values into the table → Use the query: `SELECT * FROM tablename ;` which displays the entire table.

```

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 32
Server version: 8.0.42 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE testdb;
Query OK, 1 row affected (0.029 sec)

MySQL [(none)]> USE testdb;
Database changed
MySQL [(testdb)]> CREATE TABLE students(id INT PRIMARY KEY AUTO INCREMENT,name VARCHAR(50),age INT);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax
to use near 'AUTO INCREMENT,name VARCHAR(50),age INT)' at line 1
MySQL [(testdb)]> ^C

MySQL [(testdb)]> CREATE TABLE students(id INT PRIMARY KEY AUTO_INCREMENT,name VARCHAR(50),age INT);
Query OK, 0 rows affected (0.059 sec)

MySQL [(testdb)]> INSERT INTO students(name,age) VALUES('Ayesha',15),('Anderson',20);
Query OK, 2 rows affected (0.006 sec)
Records: 2  Duplicates: 0  Warnings: 0

MySQL [(testdb)]> SELECT * FROM students;
+----+-----+-----+
| id | name  | age  |
+----+-----+-----+
| 1  | Ayesha | 15   |
| 2  | Anderson | 20   |
+----+-----+-----+
2 rows in set (0.001 sec)

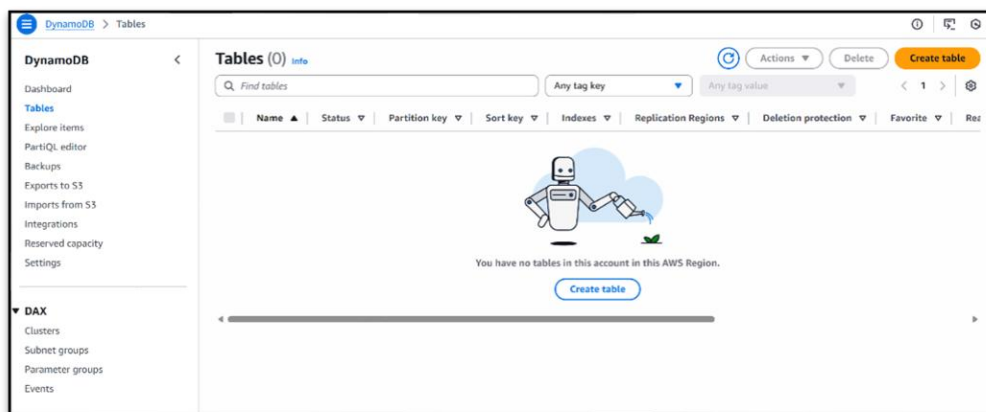
```

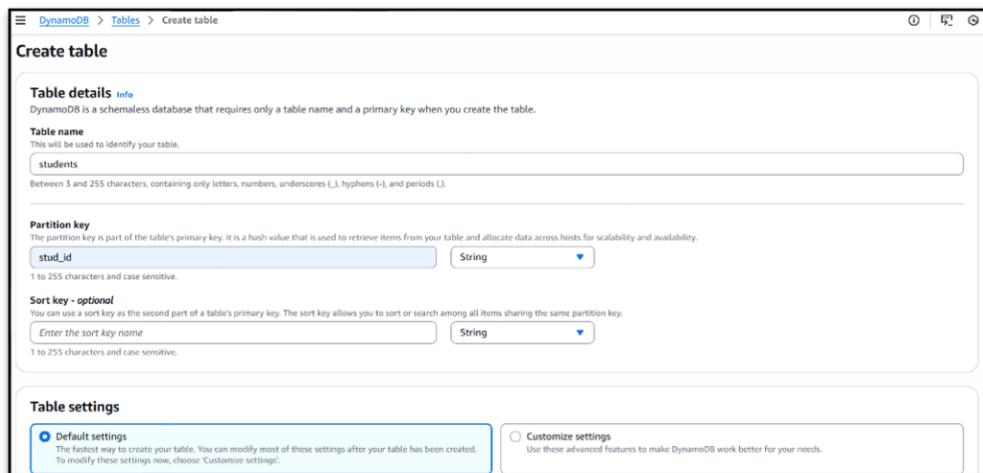
- The MySQL queries has been successfully executed using RDS and EC2 connectivity.

Task 2:

To create Dynamo DB table and write data into it.

- Open the Aws console and navigate to Dynamo DB services → Navigate to tables → Create table → Name → Partition key (It is required to have atleast one partition key → Key value → Create table.





Create table

Table details [info](#)
DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

Table name
This will be used to identify your table.
students
Between 3 and 255 characters, containing only letters, numbers, underscores (_), hyphens (-), and periods (.).

Partition key
The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.
stud_id String
1 to 255 characters and case sensitive.

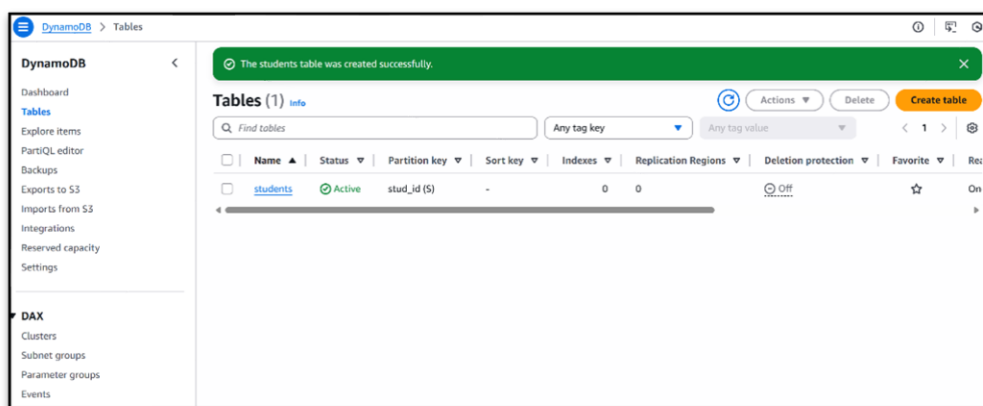
Sort key - optional
You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.
Enter the sort key name String
1 to 255 characters and case sensitive.

Table settings

☒ **Default settings**
The fastest way to create your table. You can modify most of these settings after your table has been created. To modify these settings now, choose "Customize settings".

☐ **Customize settings**
Use these advanced features to make DynamoDB work better for your needs.

- The table has been successfully created.



DynamoDB [Tables](#)

[Dashboard](#) [Tables](#) [Explore items](#) [PartiQL editor](#) [Backups](#) [Exports to S3](#) [Imports from S3](#) [Integrations](#) [Reserved capacity](#) [Settings](#)

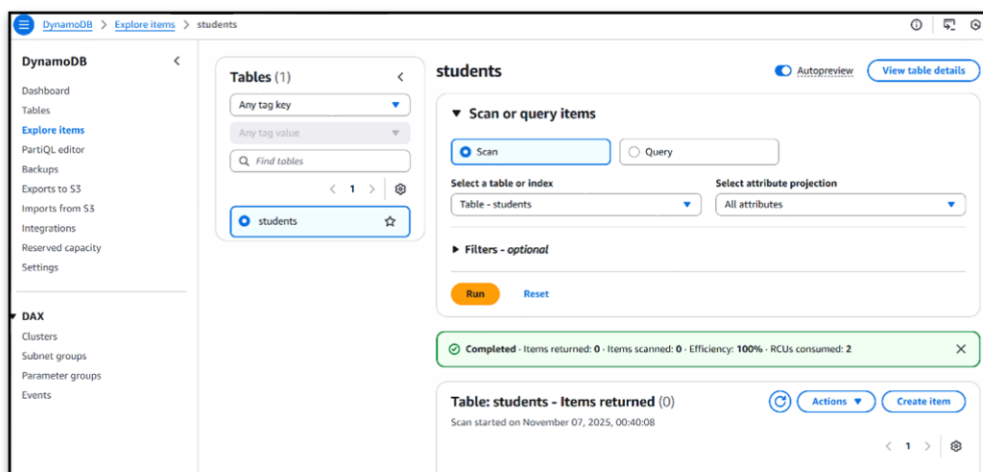
DAX [Clusters](#) [Subnet groups](#) [Parameter groups](#) [Events](#)

Tables (1) [info](#)

[Actions](#) [Delete](#) [Create table](#)

☐	Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Res
☐	students	Active	stud_id (S)	-	0	0	Off	☆	On

- Navigate to the table that has been created → Explore table items (or) directly select explore items from the bar in the left → Create item → Specify the value of the partition key and if wanted to add more keys and data → Add attributes → Select the data type (eg: String, boolean, etc.) → Create item.



DynamoDB [Explore items](#) [students](#)

[Dashboard](#) [Tables](#) [Explore items](#) [PartiQL editor](#) [Backups](#) [Exports to S3](#) [Imports from S3](#) [Integrations](#) [Reserved capacity](#) [Settings](#)

DAX [Clusters](#) [Subnet groups](#) [Parameter groups](#) [Events](#)

Tables (1) [info](#)

[students](#) ☆

students [Autopreview](#) [View table details](#)

Scan or query items

☒ Scan ☐ Query

Select a table or index: Table - students

Select attribute projection: All attributes

Filters - optional

[Run](#) [Reset](#)

Completed - Items returned: 0 - Items scanned: 0 - Efficiency: 100% - RCUs consumed: 2

Table: students - Items returned (0) [Actions](#) [Create item](#)

Scan started on November 07, 2025, 00:40:08

- If wanted to add more records the previous step is to be followed.

Create item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Attributes

Attribute name	Value	Type	Remove
stud_id - Partition key	102	String	
name	Kevin Blake	String	Remove
age	20	Number	Remove

[Add new attribute](#) [Cancel](#) [Create item](#)

- You can view the data inserted in the table.

DynamoDB

Explore items

students

Scan

Select a table or index: Table - students

Select attribute projection: All attributes

Filters - optional

[Run](#) [Reset](#)

Completed - Items returned: 2 - Items scanned: 2 - Efficiency: 100% - RCUs consumed: 2

Table: students - Items returned (2)

Scan started on November 07, 2025, 00:52:36

stud_id (String)	age	name
102	20	Kevin Blake
101	30	Rocky

- The table has been created and data has been successfully written using Dynamo DB.